Mathematical Analysis of a Three-Reel Slot Machine

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1 Introduction

Casino slot machines are among the most mathematically rich products in the gaming industry. Behind every spin lies a carefully designed probabilistic model that determines player experience through two key parameters: Return to Player (RTP) and volatility. RTP defines the expected percentage of wagered money returned to the player over a large number of spins, while volatility describes the variance of outcomes around that expected value.

This report presents the complete mathematical analysis of a simplified three-reel slot machine. The analysis covers the full workflow: theoretical derivation of key metrics, implementation of a Python-based simulation engine, statistical validation of simulated results against theoretical prediction.

The model consists of three independent reels, each sharing an identical strip of five weighted symbols. Wins are awarded exclusively for three-of-a-kind combinations. All theoretical results are derived analytically and subsequently validated through Monte Carlo simulation at scales of up to ten million spins.

2 Game Design

2.1 Reel Strip

The game consists of three identical reels, each containing 30 positions occupied by 5 symbols. Symbol weights define how many positions each symbol occupies on the strip, directly determining its probability of appearing on a given reel.

Symbol	**Weight**	**Probability**	**Payout (3-of-a-kind)**
Cherry	14	$14/30 \approx 46.67\%$	2
Lemon	9	$9/30 = 30.00\%$	4
Bell	4	$4/30 \approx 13.33\%$	10
Bar	2	$2/30 \approx 6.67\%$	30
Seven	1	$1/30 \approx 3.33\%$	100

Table 1: Reel strip composition and paytable

2.2 Win Condition

A spin wins if and only if all three reels display the same symbol (three-of-a-kind). Partial matches, scatter symbols (position-independent triggers), and wild symbols (universal substitutes) are outside the scope of this model. This deliberate simplification isolates the core combinatorial mechanics while preserving the essential mathematical structure of a real slot game.

2.3 Cost Per Spin

The cost per spin was designed to target a theoretical RTP of 98%, placing this game at the generous end of the realistic range for slot machines (typically 85%–98%). The derivation of cost per spin from target RTP is presented in Section 3.

3 Theoretical Analysis

3.1 Return to Player

Since the three reels are independent, the probability of a three-of-a-kind combination for a given symbol s with weight w_s on a strip of 30 positions is:

$$P(\text{3-of-a-kind}_s) = \left(\frac{w_s}{30}\right)^3 \tag{1}$$

Applying this to all five symbols:

Symbol	**Probability**	**Probability** $\times$ **Payout**
Cherry	$(14/30)^3 \approx 0.10163$	$0.10163 \times 2 = 0.20326$
Lemon	$(9/30)^3 = 0.02700$	$0.02700 \times 4 = 0.10800$
Bell	$(4/30)^3 \approx 0.00237$	$0.00237 \times 10 = 0.02370$
Bar	$(2/30)^3 \approx 0.00030$	$0.00030 \times 30 = 0.00888$
Seven	$(1/30)^3 \approx 0.000037$	$0.000037 \times 100 = 0.00370$

Table 2: Per-symbol win probabilities and expected payouts

The raw expected payout per spin (before cost adjustment) is:

$$E[X] = \sum_s P(\text{3-of-a-kind}_s) \times \text{payout}_s \approx 0.3476 \tag{2}$$

The cost per spin c is derived from the target RTP of 98%:

$$\text{RTP} = \frac{E[X]}{c} \implies c = \frac{E[X]}{\text{RTP}} = \frac{0.3476}{0.98} \approx 0.3546 \tag{3}$$

3.2 Hit Frequency

The hit frequency is the probability of landing any winning combination:

$$P(\text{win}) = \sum_s P(\text{3-of-a-kind}_s) \approx 0.1316 \tag{4}$$

This means approximately 13.16% of spins result in a win of some kind.

3.3 Variance and Volatility

Volatility is quantified through the variance of the payout distribution. Using the identity $\text{Var}(X) = E[X^2] - (E[X])^2$:

$$E[X^2] = \sum_s P(\text{3-of-a-kind}_s) \times \text{payout}_s^2 \approx 1.7119 \tag{5}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 1.7119 - (0.3476)^2 \approx 1.5910 \tag{6}$$

$$\sigma = \sqrt{\text{Var}(X)} \approx 1.2614 \tag{7}$$

The standard deviation $\sigma \approx 1.26$ is approximately 3.6 times the expected payout per spin, indicating a high-volatility game where most spins yield no return but occasional wins are disproportionately large.

4 Simulation

4.1 Implementation

The simulation engine is implemented in Python using an object-oriented approach. The `SlotMachine` class encapsulates the game parameters and exposes two methods: `spin()`, which simulates a single spin, and `run_simulation()`, which runs n spins and returns aggregated statistics.

Weighted symbol selection is implemented by constructing a reel strip list where each symbol appears a number of times equal to its weight. Python's `random.choice()` function then selects uniformly at random from this list, naturally reproducing the desired probability distribution without requiring explicit weighted sampling.

4.2 Core Algorithm

A single spin proceeds as follows:

1. Sample one symbol independently from each of the three reels using `random.choice()`
2. Check if all three symbols match (three-of-a-kind condition)
3. Return the corresponding payout from the paytable dictionary, or zero if no match

The following code excerpt shows the core spin logic:

```
def spin(self):
    reel1 = random.choice(self.reel_strip)
    reel2 = random.choice(self.reel_strip)
    reel3 = random.choice(self.reel_strip)
    if reel1 == reel2 and reel1 == reel3:
        return (f"{reel1}-{reel2}-{reel3}", self.payout_table[reel1])
    else:
        return (f"{reel1}-{reel2}-{reel3}", 0)
```

4.3 Output

The simulation produces two CSV files:

- `three_reel_slot_simulation.csv` — spin-level data (spin number, symbol combination, first reel symbol, payout, win flag) used for Excel-based validation and visualization
- `convergence.csv` — cumulative RTP at every 1,000 spin checkpoint, used to visualize convergence behavior

4.4 Playable Mode

In addition to bulk simulation, the engine supports an interactive play mode where the user enters a starting balance and spins one at a time, observing the symbol combination and net result of each spin. This mode demonstrates the game experience directly and provides an intuitive feel for the volatility of the design.

5 Results

5.1 Simulation Validation

Table 3 compares theoretical predictions against simulated results at 10,000,000 spins. All four metrics show excellent agreement, confirming the correctness of both the theoretical model and the simulation engine.

Metric	Theoretical	Simulated	Difference
RTP	98.00%	97.95%	−0.05%
Hit Frequency	13.16%	13.12%	−0.04%
Variance	1.5911	1.5925	+0.0014
Std Deviation	1.2614	1.2620	+0.0006

Table 3: Theoretical vs simulated metrics at 10,000,000 spins

5.2 Symbol Frequency Validation

Figure 1 shows the simulated symbol frequency on Reel 1 compared to the theoretical probability for each symbol. All five symbols show agreement within 0.15 percentage points of theory, confirming the reel strip weights are correctly implemented.

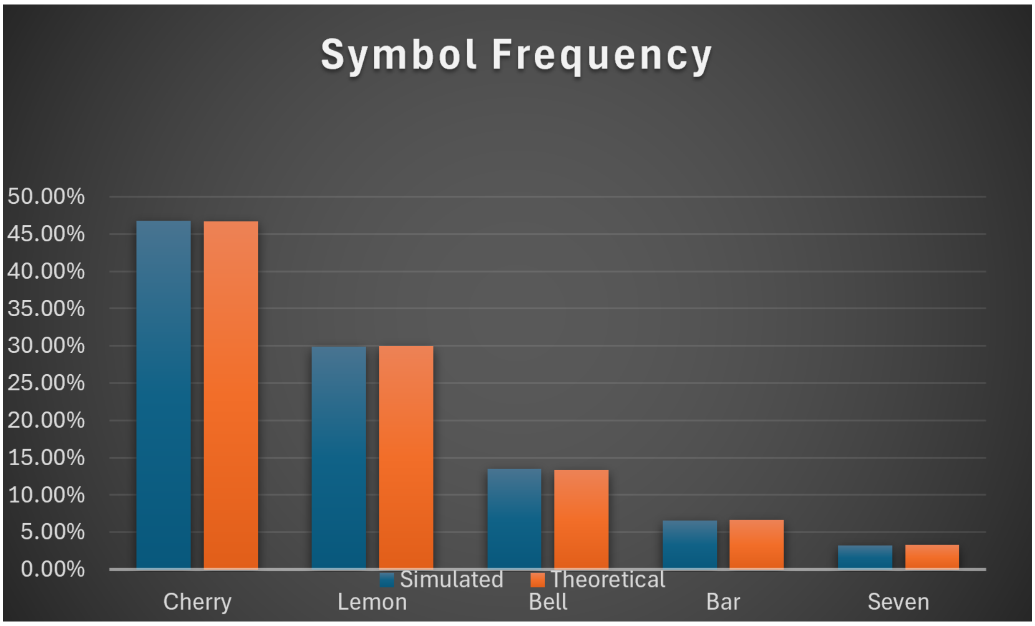

Figure 1: Simulated vs theoretical symbol frequency — Reel 1, 100,000 spins

5.3 Payout Distribution

Figure 2 shows the frequency of each payout value across 100,000 spins. The distribution is heavily right-skewed: 86.78% of spins yield no payout, while large wins (Bar: 30 credits, Seven: 100 credits) occur with near-zero frequency at this sample size. This shape is characteristic of a high-volatility game design.

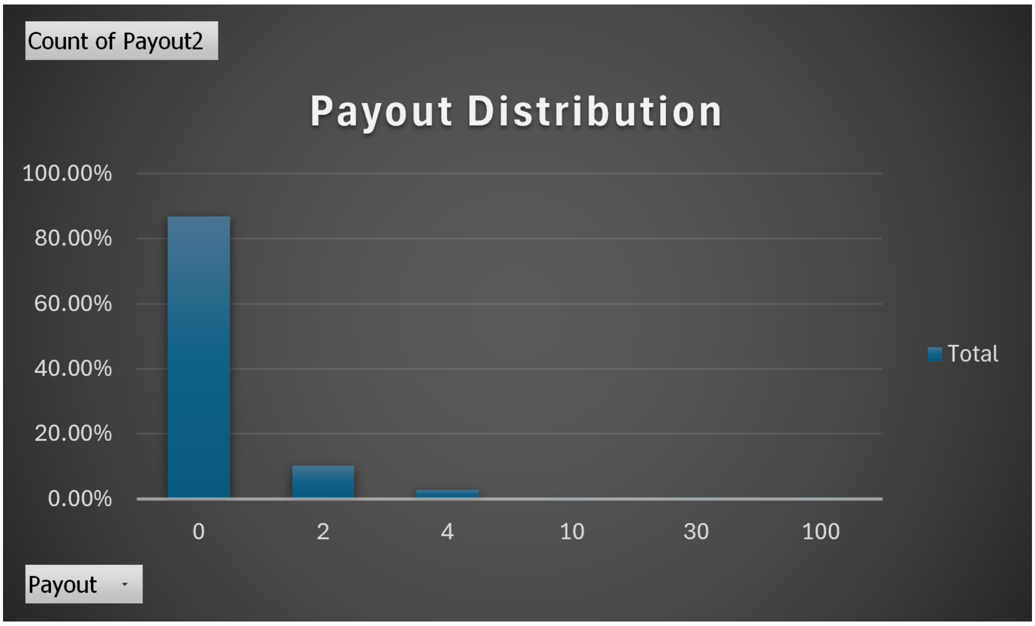

Figure 2: Payout frequency distribution — 100,000 spins

5.4 RTP Convergence

Figure 3 shows the cumulative RTP over 1,000,000 spins, sampled at 1,000 spin intervals. The theoretical target of 98% is shown as a horizontal reference line. The simulated RTP exhibits high volatility in early spins before converging toward the theoretical value, consistent with the law of large numbers.

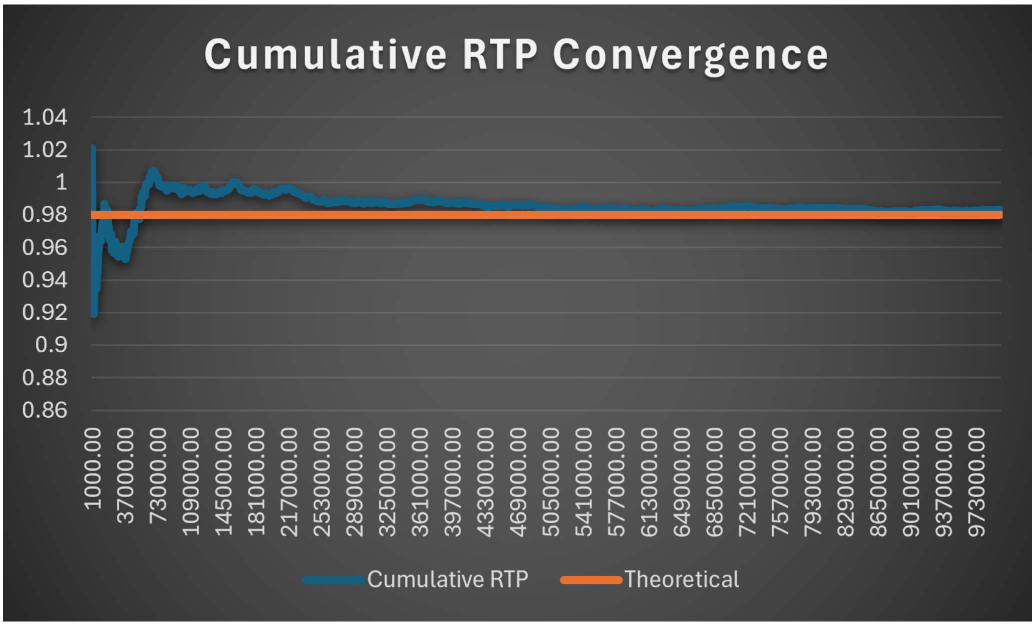

Figure 3: Cumulative RTP convergence toward theoretical 98% target

6 Convergence Analysis

A key observation from the simulation results is that different metrics converge to their theoretical values at different rates. Table 4 summarizes the simulated values at three sample sizes for each metric.

Metric	**Theoretical**	**10,000 spins**	**100,000 spins**	**10,000,000 spins**
RTP	98.00%	98.87%	~95–101%	97.95%
Hit Frequency	13.16%	13.25%	13.17%	13.12%
Variance	1.5911	1.2494	~1.25–1.67	1.5925

Table 4: Metric convergence across sample sizes

Hit frequency converges rapidly because it depends only on whether a spin wins or loses — a binary outcome with relatively high probability (13.16%). By contrast, RTP and variance converge slowly because they are dominated by rare, high-payout events. To see why, consider the contribution of the Seven symbol to $E[X^2]$:

$$P(\text{Three Sevens}) \times \text{payout}^2 = 0.000037 \times 100^2 = 0.37 \tag{8}$$

This single symbol contributes 0.37 out of the total $E[X^2] = 1.7119$, approximately 21.6% of the total — despite having a probability of only 0.0037%. At 10,000 spins, Seven is expected to appear only $0.000037 \times 10{,}000 \approx 0.37$ times on average, meaning it frequently does not appear at all, causing $E[X^2]$ and therefore variance to be severely underestimated.

This phenomenon is well known in Monte Carlo simulation: estimating higher-order moments (variance, skewness) requires substantially larger sample sizes than estimating the mean, particularly for heavy-tailed distributions where rare large values dominate the moment calculation.

In practice, professional game mathematics validation uses sample sizes of 10^7 to 10^9 spins precisely for this reason — smaller samples produce unreliable variance estimates even when RTP appears converged.

7 Conclusion

This report presented a complete mathematical analysis of a three-reel slot machine, covering game design, theoretical derivation, simulation implementation, and statistical validation.

The key results are as follows:

- Theoretical RTP of 98.00% was confirmed by simulation to within 0.05% at 10,000,000 spins
- Hit frequency of 13.16% was confirmed to within 0.04%
- Variance and standard deviation converged more slowly due to the disproportionate contribution of rare high-payout symbols to $E[X^2]$
- The payout distribution exhibits strong right skew, characteristic of a high-volatility game design

7.1 Limitations

The current model makes several simplifying assumptions:

- All three reels share an identical strip — real slot machines typically vary reel weights to independently tune RTP and hit frequency
- Wins are awarded only for three-of-a-kind combinations — real games include partial wins, scatter triggers, and wild symbols
- The model contains no bonus features or progressive jackpot mechanics

7.2 Future Work

Several natural extensions would increase the realism and mathematical complexity of this model:

- **Non-identical reels:** allowing each reel to have independent symbol weights, requiring per-reel probability calculations and enabling finer RTP tuning

- **Conditional probability mechanics:** implementing a bonus trigger where landing a specific symbol on Reel 1 modifies the probability distribution of Reel 2, introducing statistical dependence between reels and requiring conditional probability and Bayes' theorem for theoretical analysis
- **Wild and scatter symbols:** extending the win condition to include substitution and position-independent triggers
- **Larger simulation scale:** running 10^8–10^9 spins to achieve tighter variance convergence

References

[1] Milan Merkle, *Verovatnoća i statistika za inženjere i studente tehnike*, 4th edition, Akademska misao, 2016.